

# TCS349: Responsible and Explainable AI

## Unit 1, Lecture 5

### Algorithmic Bias, Interactional Bias, Feedback Loops and Deployment Bias

Today we will see how bias can enter even after data collection — through algorithms, user interaction, repeated feedback and real-world deployment.

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## How Bias Can Enter After Data Collection

*The AI lifecycle does not end after we prepare data.*



*Lecture 4 mostly focused here.*

***Lecture 5 focuses here.***



**Feedback loop develops when outputs become future inputs.**

## Warm-Up: A Model Can Become Biased Even with Good Data

Imagine we trained a model using a balanced dataset.

The model is accurate overall, but one group still receives more false rejections.

Question: Is the dataset always responsible for the problem?

Answer: Not always. Bias can also come from algorithms, thresholds, interaction, feedback or deployment conditions.



# 1. Algorithmic Bias

Algorithmic bias arises from the model design, objective function, optimisation method, feature use or decision threshold.

The data may be imperfect, but the algorithm can further amplify unequal treatment.

A model can optimise overall accuracy while making more errors for smaller groups.

A single threshold may work well for the majority but poorly for minority subgroups.



## Example: Credit Scoring Threshold

A bank model predicts whether a loan applicant will repay a loan.

The bank sets one common approval **threshold** for everyone.

Overall accuracy looks good, but false rejection is higher for one subgroup.

Ethical concern: the model may deny opportunity to people who were actually creditworthy.

**Teaching question:** Should we use one threshold for everyone, or group-aware fairness checks?



## 2. Interactional Bias

Interactional bias arises from the interaction between users and the AI system.

The system influences user behaviour; user behaviour then influences what the system learns.

The bias may not be present only in the original data or model. It emerges during use.

Examples: search autocomplete, recommender systems, social media feeds and educational platforms.



## Example: Search and Recommendation Interaction



A platform recommends a certain type of content more often.

Users click on what is shown because it is easily visible.

The model learns that this content is more preferred.

The system recommends similar content even more strongly next time.

Result: user choices may become narrower over time.



### 3. Feedback Loops

A feedback loop occurs when the output of an AI system affects the future data used to train or update the same system.

This can magnify small differences into large unfair outcomes.

Positive feedback loop: the system reinforces what it already favours.

Negative outcome: alternatives, minority groups or rare cases become less visible.





## Feedback Loop Case: Predictive Policing

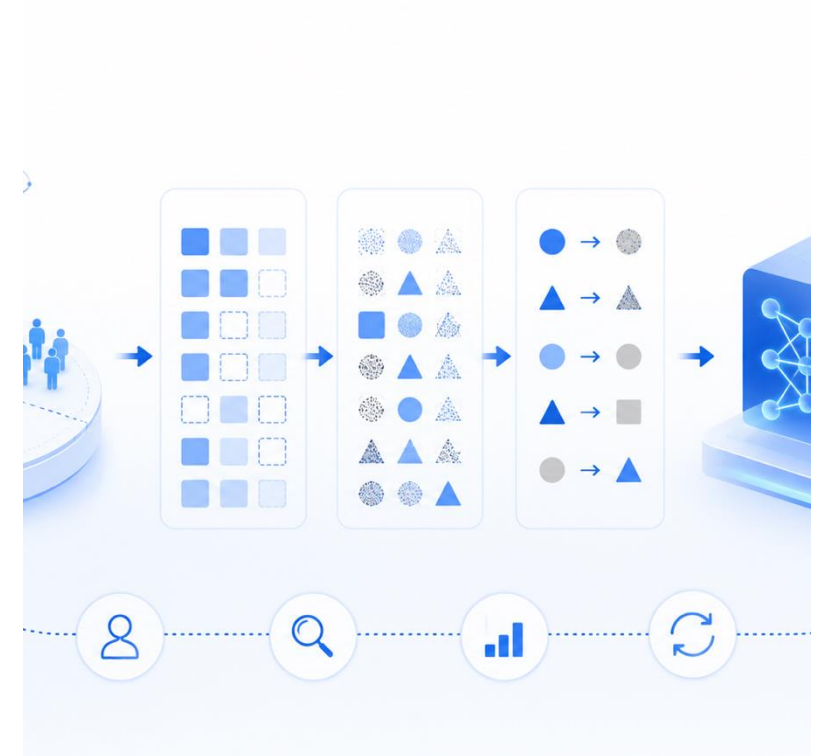
If an algorithm predicts more crime in one area, more policing may be sent there.

More policing may detect more incidents in that area, even if crime is not actually higher overall.

The new data then confirms the original prediction.

The loop can reinforce surveillance of the same communities.

Ethical concern: the system may confuse “more observed incidents” with “more actual risk.”



## Feedback Loop Case: Popularity Bias in Recommenders



Popular items receive more recommendations.

More recommendations create more clicks and views.

More clicks make the items look even more popular.

New or niche content may remain invisible even if it is high quality.

This can reduce diversity and reinforce dominant trends.

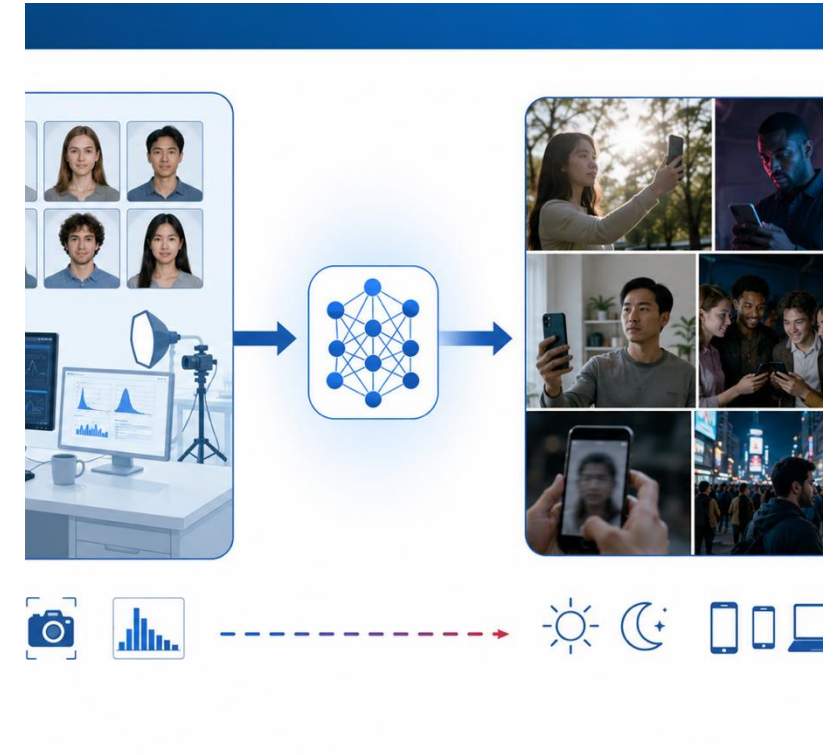
## 4. Deployment Bias

Deployment bias occurs when an AI system is used in a context different from the one for which it was designed, trained or tested.

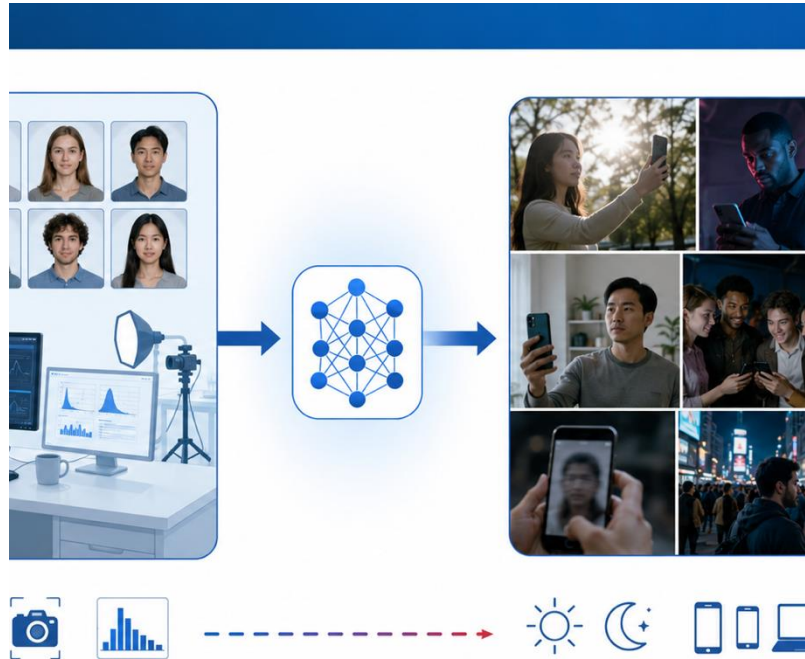
The model may perform well in the lab but fail in real-world conditions.

The user population, environment, devices, lighting, language or behaviour may change.

Responsible AI requires context-aware testing before deployment.



## Deployment Bias Example: Face Recognition in Real Conditions



A system may be trained on clear, frontal, high-quality images.

In real use, faces may appear in poor lighting, crowd scenes, CCTV angles or mobile cameras.

Accuracy may drop for certain groups or conditions.

A model validated in one country, campus or environment may not be reliable everywhere.

Key lesson: model evaluation must match real deployment conditions.

## Quick Comparison: Four Biases in Today's Lecture

| Bias Type     | Where It Arises                                           | Simple Example                                         |
|---------------|-----------------------------------------------------------|--------------------------------------------------------|
| Algorithmic   | Model design, objective, threshold or optimisation choice | Minority group gets more errors after training         |
| Interactional | How users interact with the AI system                     | Users change behaviour because of recommendations      |
| Feedback loop | Model output changes future data                          | Popular content becomes even more popular              |
| Deployment    | Mismatch between training context and real use            | Model works in lab but fails in messy field conditions |

## Case Analysis: AI Hiring Tool

### Where Bias May Enter

Algorithmic bias: model weighs certain features too strongly

Interactional bias: recruiters over-trust the model score

Feedback loop: selected candidates shape future training data

Deployment bias: tool built for one industry used in another context

### Questions for Students

Who is affected by the model output?

What decision is being automated or supported?

Which bias type is most visible here?

What safeguard would you add before deployment?